

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T55 · Cumulative Frequency

Questions

Student book · questions with writing space

■ 34 pages

FAMILY MAP

Cumulative Frequency Diagrams

F01 Construct cumulative-frequency diagrams 5 Q | 18 marks

F02 Read counts and thresholds from cumulative frequency 9 Q | 42 marks

F03 Estimate and compare location and spread from cumulative frequency 9 Q | 19 marks

- 13** The frequency table gives information about the times, in minutes, that 60 people took to complete a puzzle.

Time (t minutes)	Frequency
$0 < t \leq 10$	8
$10 < t \leq 20$	13
$20 < t \leq 30$	12
$30 < t \leq 40$	17
$40 < t \leq 50$	7
$50 < t \leq 60$	3

- (a) Complete the cumulative frequency table.

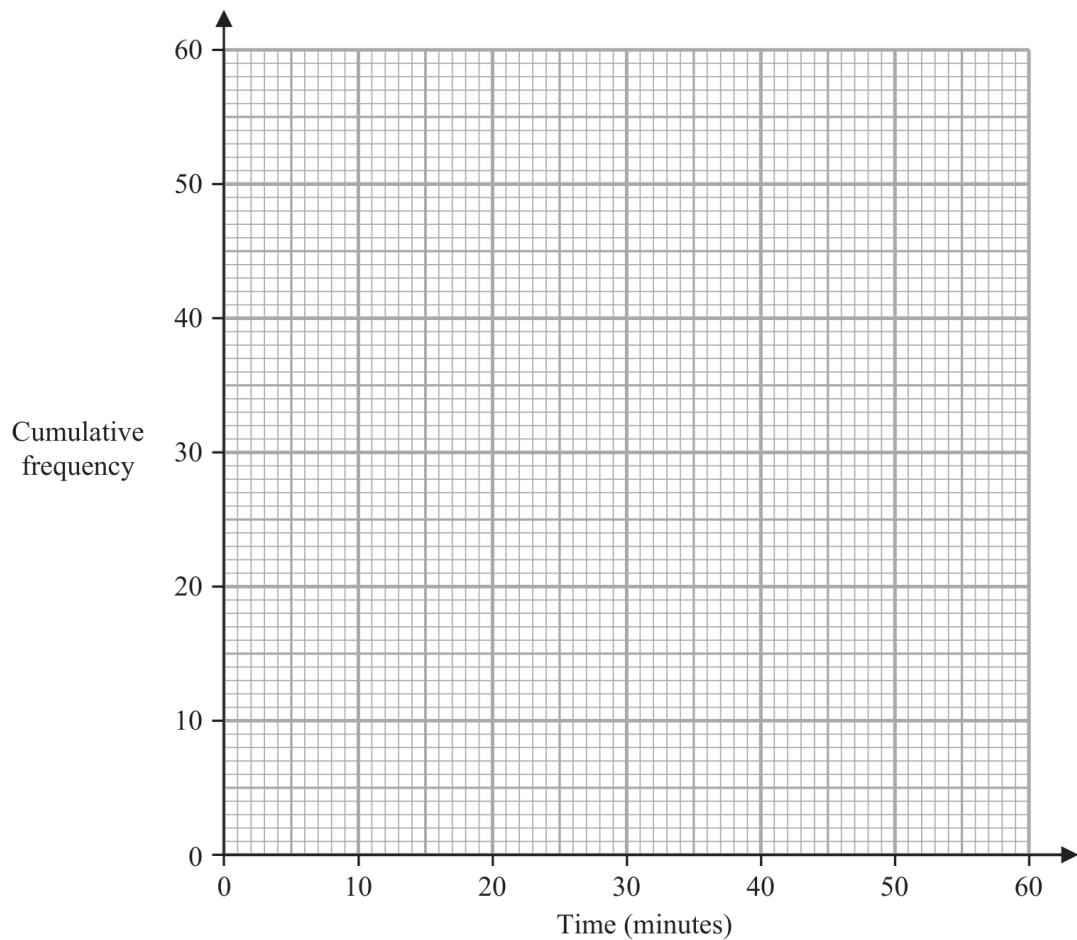
Time (t minutes)	Cumulative frequency
$0 < t \leq 10$	
$0 < t \leq 20$	
$0 < t \leq 30$	
$0 < t \leq 40$	
$0 < t \leq 50$	
$0 < t \leq 60$	

(1)

- (b) On the grid opposite, draw a cumulative frequency graph for your table.

Continued
4MA1-2H Q13

UNIT 2



(2)

- (c) Use your graph to find an estimate for the median time.

..... minutes
(1)

- (d) Use your graph to find an estimate for the number of these people who took more than 45 minutes to complete the puzzle.

Question 18

4MA1-2H Q13

MODULAR UNIT 2

7 marks

13 The table gives information about the distances, in km, that 70 teachers travel to school.

Distance (d km)	Frequency
$0 < d \leq 10$	7
$10 < d \leq 20$	17
$20 < d \leq 30$	18
$30 < d \leq 40$	14
$40 < d \leq 50$	10
$50 < d \leq 60$	4

(a) Complete the cumulative frequency table.

Distance (d km)	Cumulative frequency
$0 < d \leq 10$	
$0 < d \leq 20$	
$0 < d \leq 30$	
$0 < d \leq 40$	
$0 < d \leq 50$	
$0 < d \leq 60$	

Question 18 (continued)

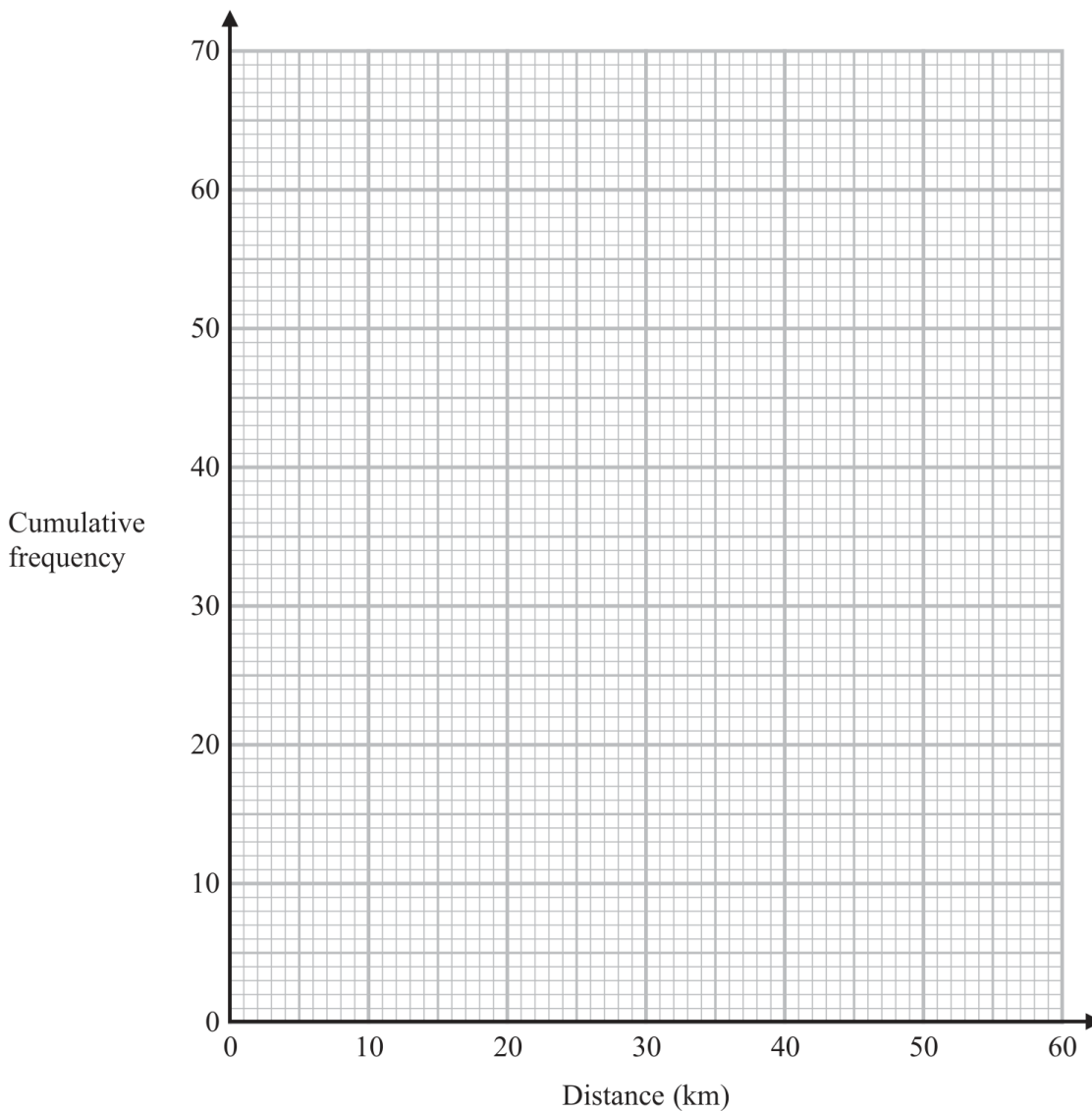
4MA1-2H Q13

MODULAR UNIT 2

7 marks

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the interquartile range of the distances.

..... km

(2)

4MA1-2H Q13

7 marks

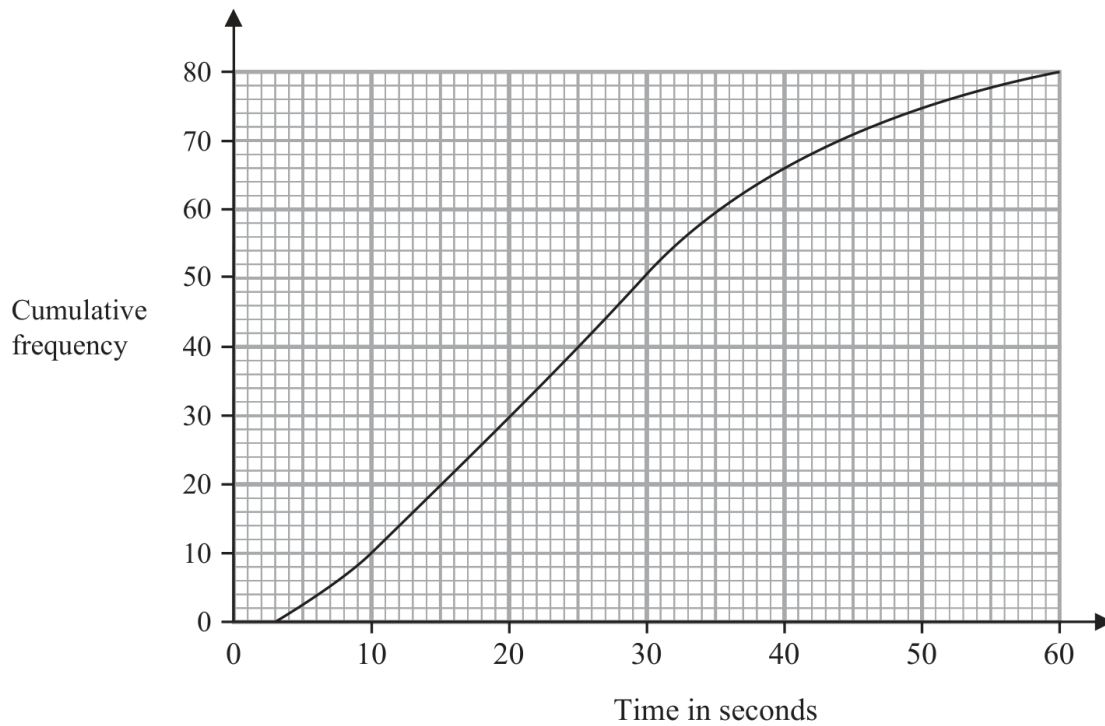
Question 09

4MA1-1H Q13 (a)

MODULAR UNIT 2

1 marks

- 13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



- (a) Find an estimate for the median time.

..... seconds

(1)

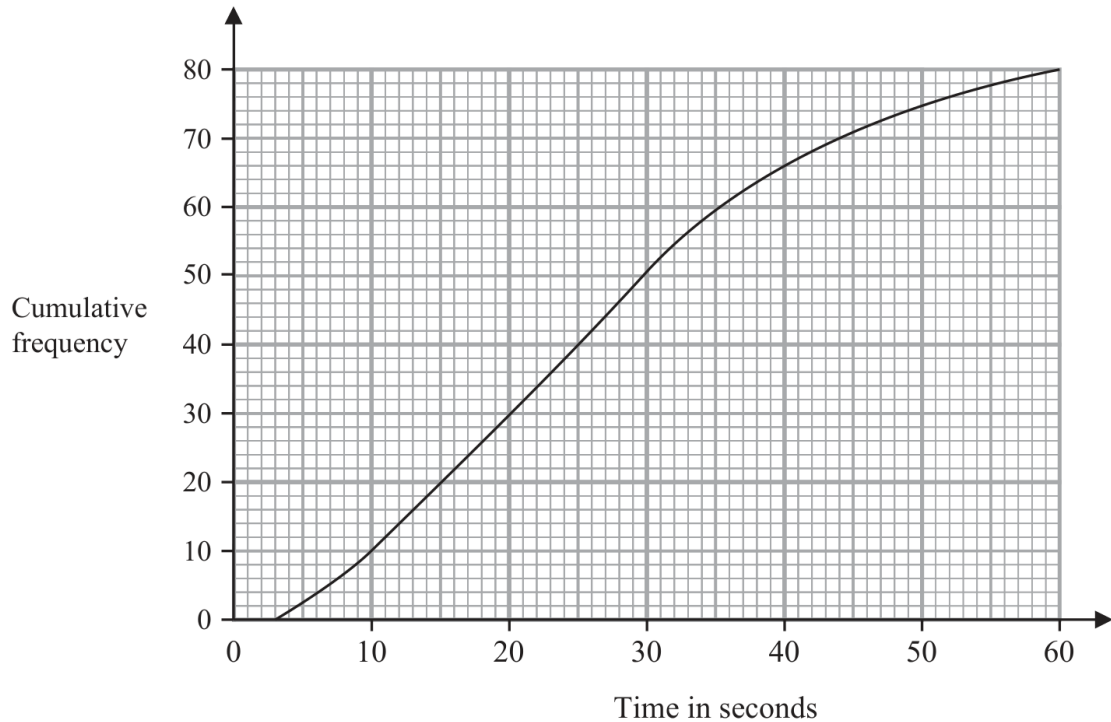
Question 10

4MA1-1H Q13 (b)

MODULAR UNIT 2

3 marks

- 13 The cumulative frequency graph gives information about the times, in seconds, that 80 adults took to log in to an online bank.



- (b) Work out the percentage of these adults that took longer than 50 seconds to log in.
Show your working clearly.

..... %
(3)

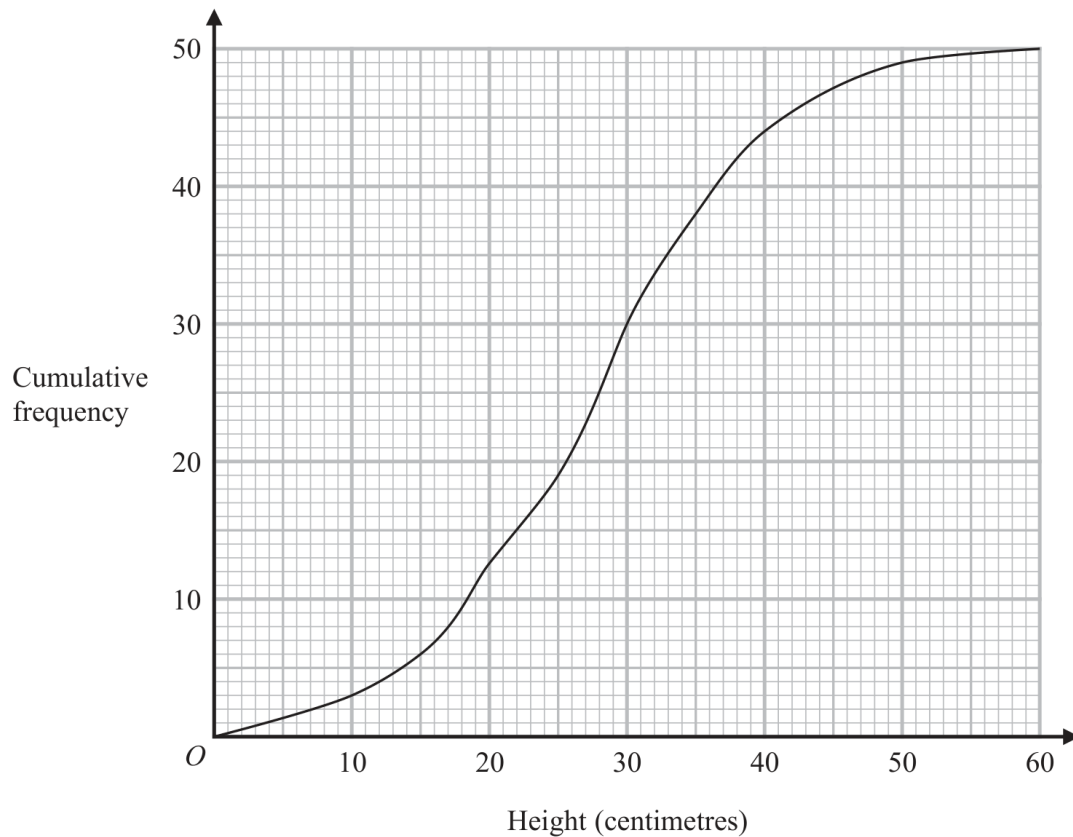
Question 12

4MA1-1H Q12

MODULAR UNIT 2

4 marks

- 12 The cumulative frequency graph shows information about the heights, in centimetres, of 50 plants in a flowerbed.



- (a) Use the graph to find an estimate for the median height of these plants.

..... centimetres
(1)

- (b) Use the graph to find the frequency for the class interval $30 < \text{Height} \leq 40$

.....
(1)

- (c) Use the graph to find an estimate for the number of plants with a height greater than 35 centimetres.

Question 27

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2

6 marks

- 13 The frequency table gives information about the weights, in kilograms, of 60 parcels in a delivery van.

Weight (w kilograms)	Frequency
$0 < w \leq 1$	4
$1 < w \leq 2$	15
$2 < w \leq 3$	20
$3 < w \leq 4$	11
$4 < w \leq 5$	6
$5 < w \leq 6$	4

- (a) Complete the cumulative frequency table.

Weight (w kilograms)	Cumulative frequency
$0 < w \leq 1$	
$0 < w \leq 2$	
$0 < w \leq 3$	
$0 < w \leq 4$	
$0 < w \leq 5$	
$0 < w \leq 6$	

(1)

- (b) On the grid opposite, draw a cumulative frequency graph for your table.

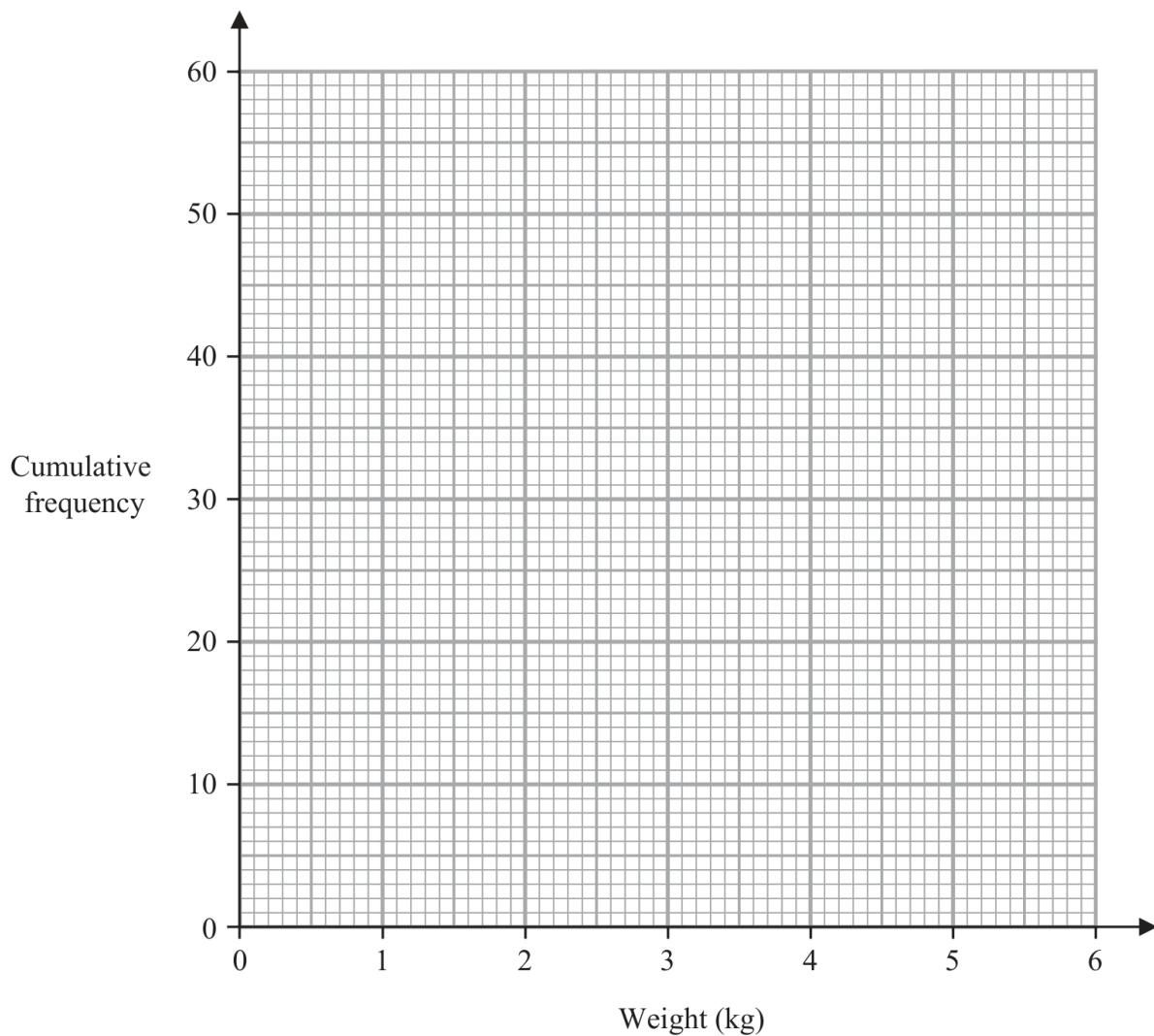
(2)

Question 27 (continued)

4MA1-2H Q13 M01, M02, M03, M04

MODULAR UNIT 2

6 marks



- (c) Use your graph to find an estimate for the median weight of the 60 parcels.

..... kilograms
(1)

- (d) Use your graph to find an estimate for the number of these parcels that weigh more than 3.7 kilograms.

Question 26

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

MODULAR UNIT 2

7 marks

13 The table gives information about the ages, in years, of 80 people in a train carriage.

Age (a years)	Frequency
$0 < a \leq 20$	7
$20 < a \leq 30$	25
$30 < a \leq 40$	20
$40 < a \leq 50$	14
$50 < a \leq 60$	8
$60 < a \leq 70$	6

(a) Complete the cumulative frequency table.

Age (a years)	Cumulative frequency
$0 < a \leq 20$	
$0 < a \leq 30$	
$0 < a \leq 40$	
$0 < a \leq 50$	
$0 < a \leq 60$	
$0 < a \leq 70$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your table.

Question 26 (continued)

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

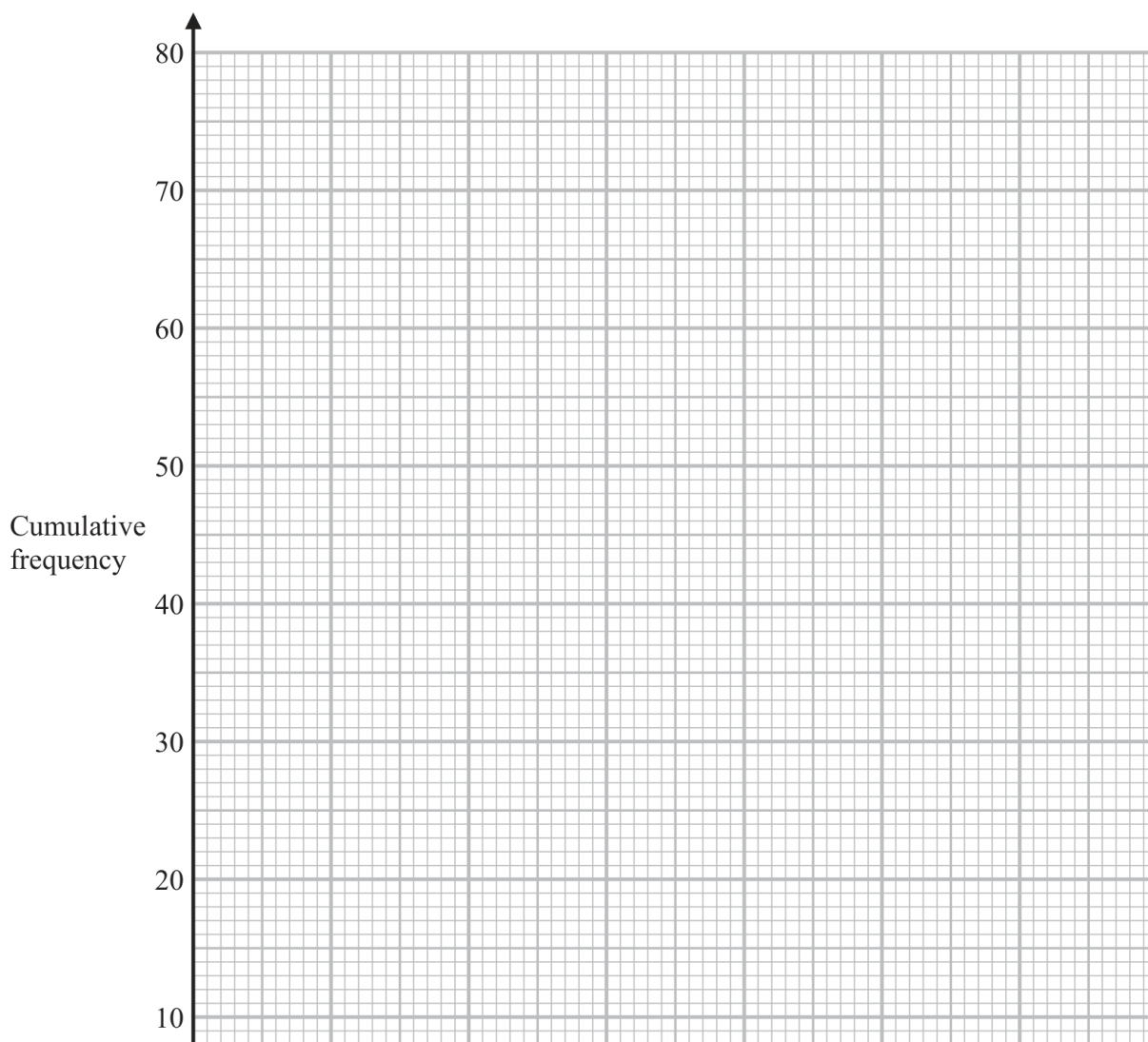
MODULAR UNIT 2**7 marks**

(2)

(c) Use your graph to find an estimate for the median age of the 80 people.

..... years

(1)

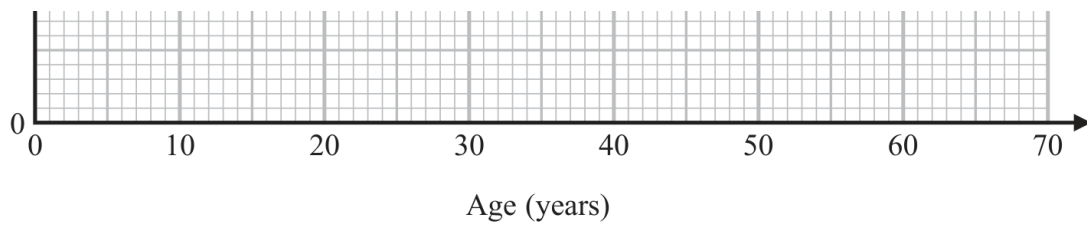


Question 26 (continued)

4MA1-2H Q13 Q13(a), Q13(b), Q13(c), Q13(d)

MODULAR UNIT 2

7 marks



Of the people in the train carriage, 60% of those who are aged between 18 and 65 are going to work. None of the other people in the train carriage are going to work.

- (d) Use your graph to find an estimate for the number of people in the train carriage who are going to work.

MODULAR UNIT 2

6 marks

- | Time taken (t minutes) | Frequency |
|---------------------------|-----------|
| $0 < t \leq 10$ | 7 |
| $10 < t \leq 20$ | 26 |
| $20 < t \leq 30$ | 24 |
| $30 < t \leq 40$ | 14 |
| $40 < t \leq 50$ | 7 |
| $50 < t \leq 60$ | 2 |

- | Time taken (t minutes) | Cumulative frequency |
|---------------------------|----------------------|
| $0 < t \leq 10$ | |
| $0 < t \leq 20$ | |
| $0 < t \leq 30$ | |
| $0 < t \leq 40$ | |
| $0 < t \leq 50$ | |
| $0 < t \leq 60$ | |

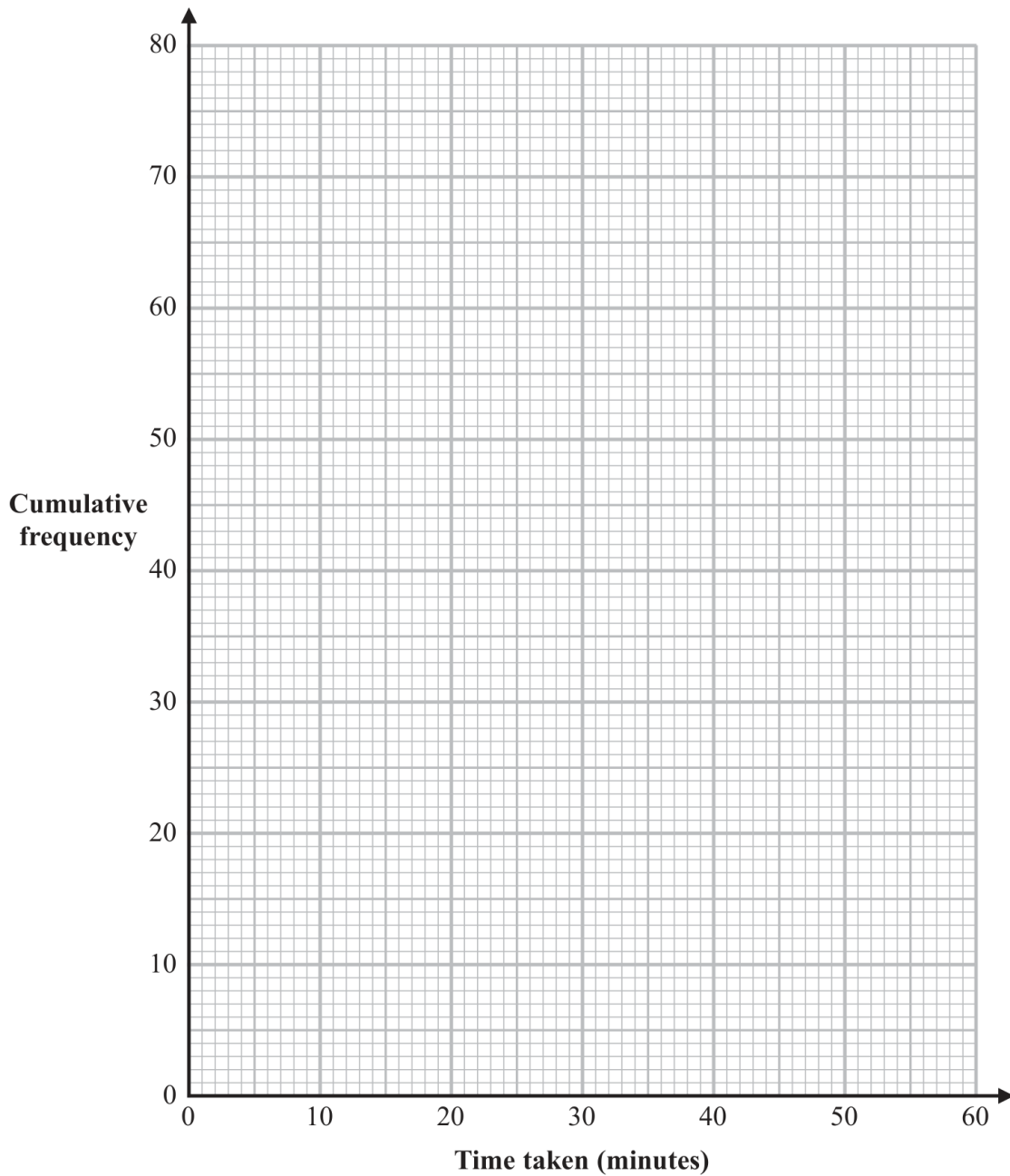
(b) On the grid opposite, draw a cumulative frequency graph for your table.

Question 09 (continued)

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

MODULAR UNIT 2

6 marks



(2)

Question 09 (continued)

4MA1-1H Q12 Q12(a), Q12(b), Q12(c), Q12(d)

MODULAR UNIT 2

6 marks

(c) Use your graph to find an estimate for the median time taken.

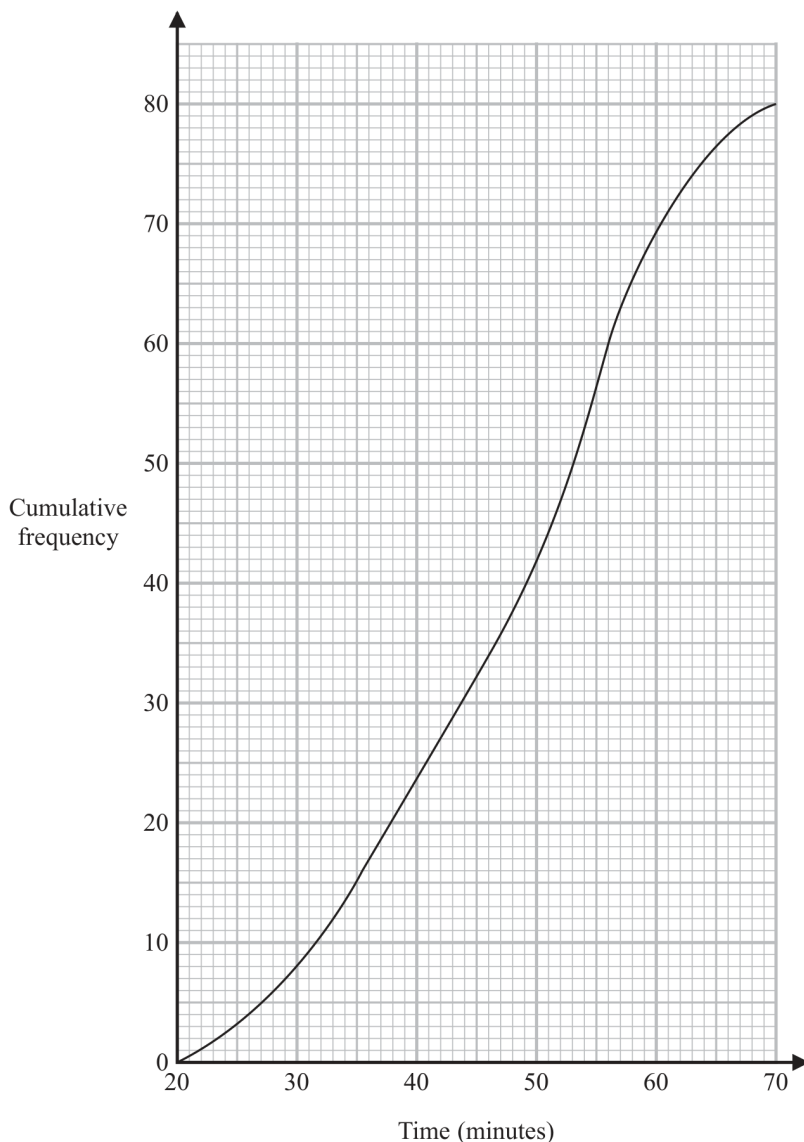
..... minutes

(1)

One of the 80 customers is chosen at random.

(d) Use your graph to find an estimate for the probability that the time taken by this customer was more than 42 minutes.

- 12 A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.



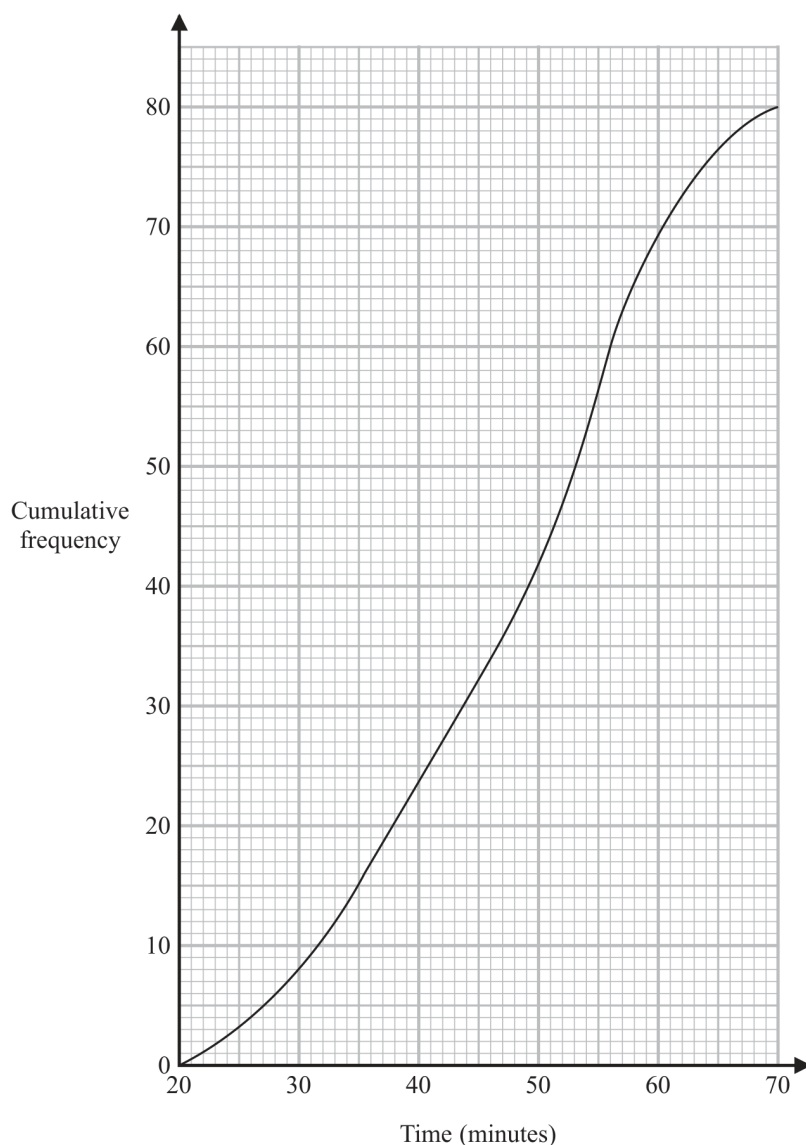
- (a) Use the graph to find an estimate for the interquartile range.

..... minutes
(2)

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

- (b) Work out an estimate for the number of men who took part in the race.

- 12 A total of 80 men and women took part in a race.
The cumulative frequency graph gives information about the times, in minutes, they took for the race.



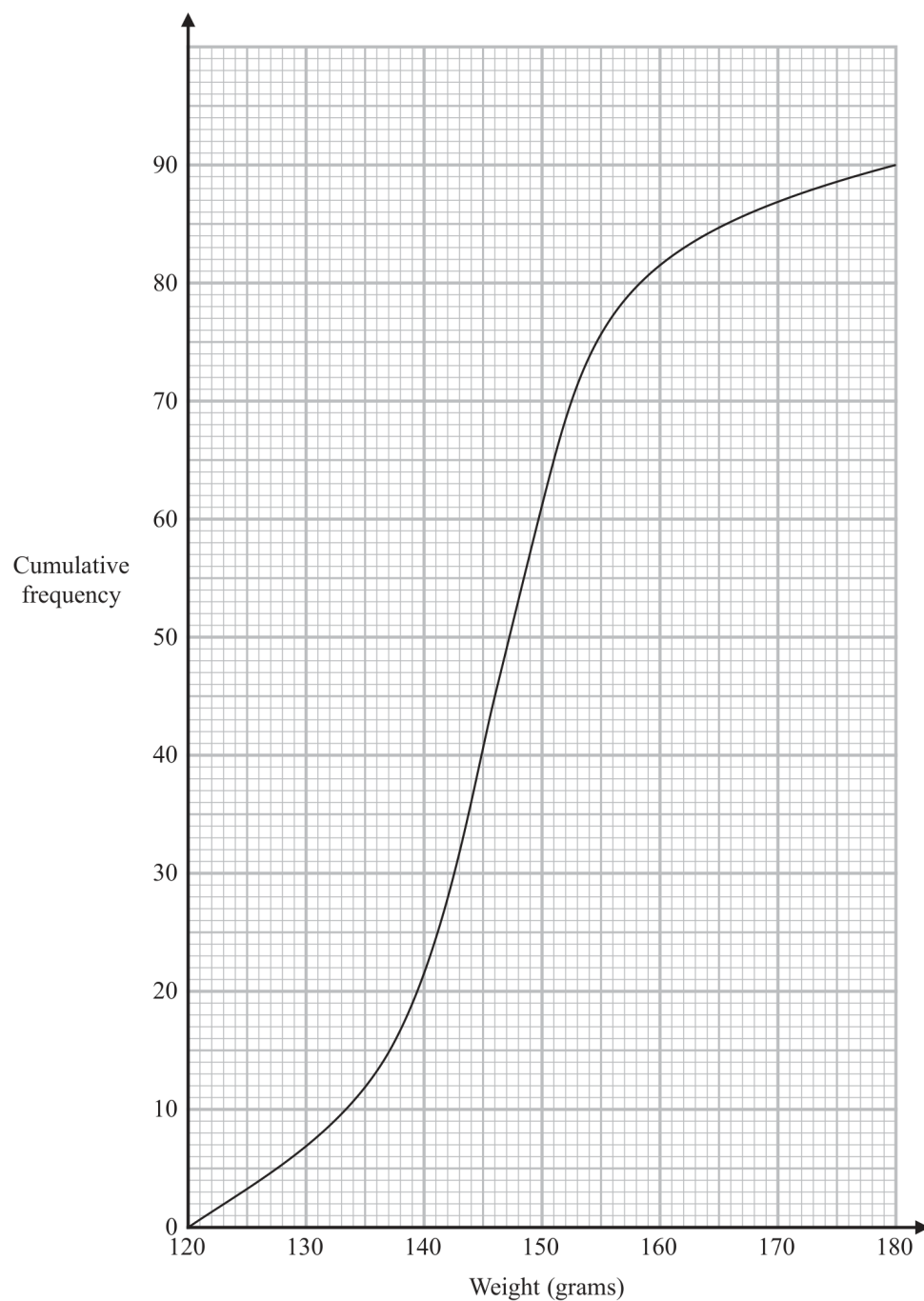
- (a) Use the graph to find an estimate for the interquartile range.

..... minutes
(2)

60% of the men took 50 minutes or less for the race.
No women took 50 minutes or less for the race.

- (b) Work out an estimate for the number of men who took part in the race.

- 11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets.



- (a) Find an estimate for the median of the weights of these bags of sweets.

Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag.

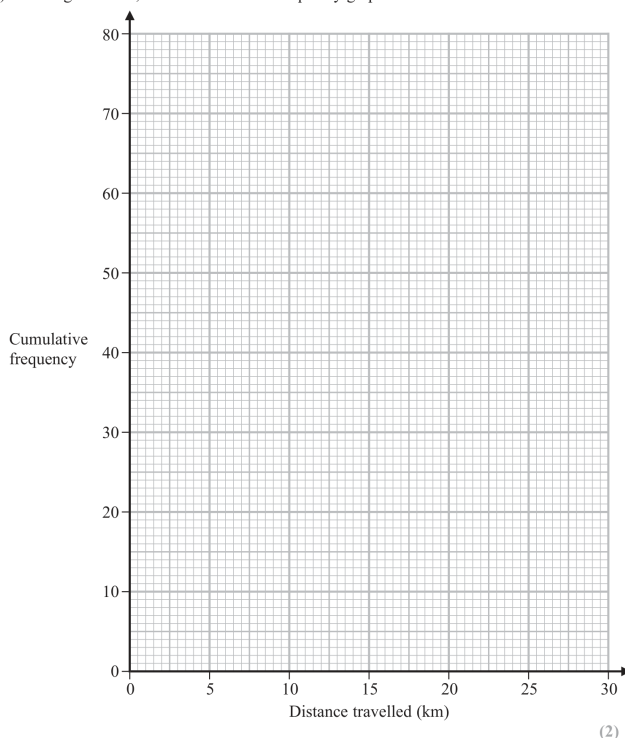
(b) Find the value of d .

$$d = \dots\dots\dots (3)$$

- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km
(1)

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km
(2)

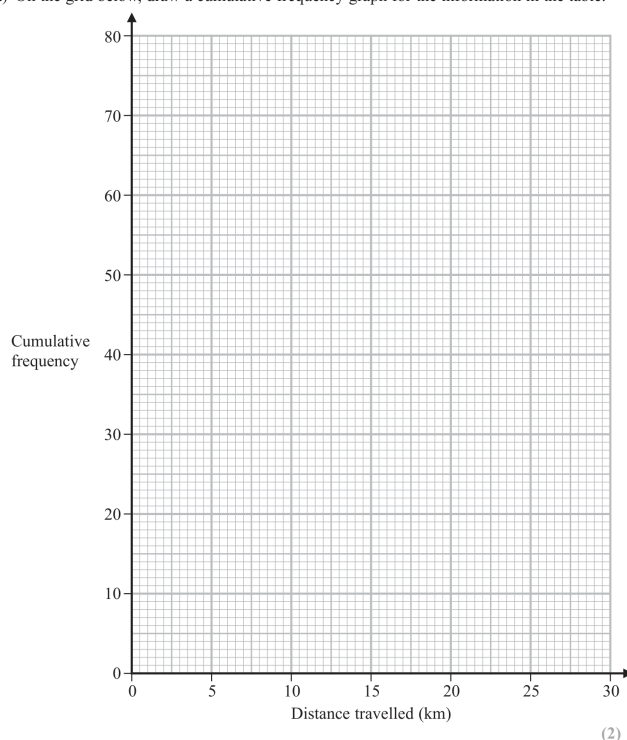
For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km
(1)

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km
(2)

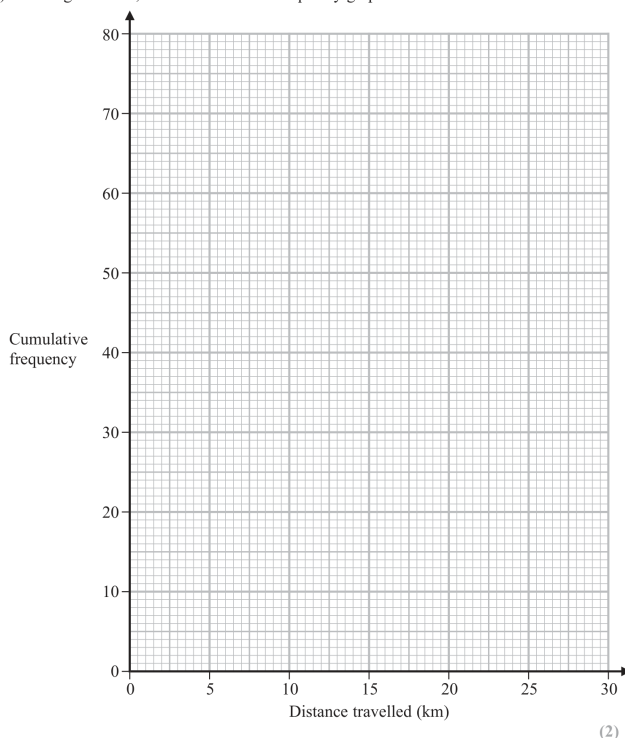
For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km
(1)

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km
(2)

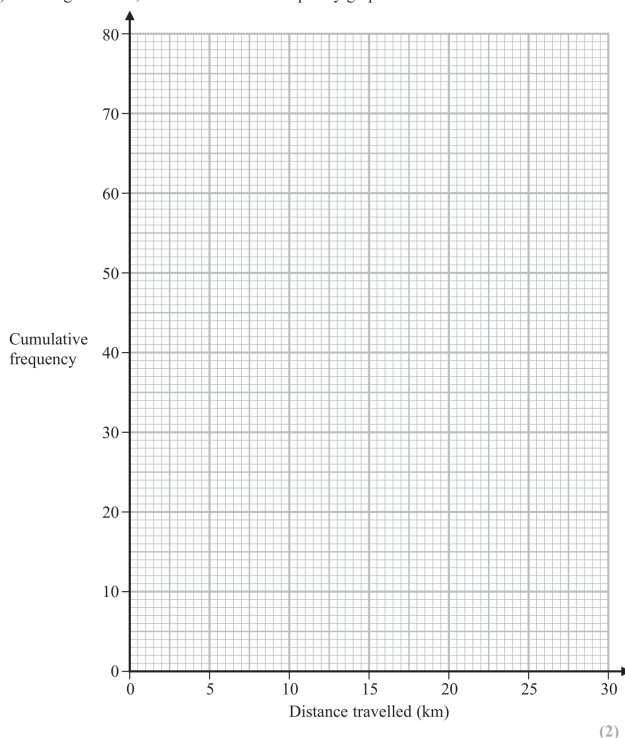
For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

- 12 The cumulative frequency table gives information about the distance, in kilometres, that each of 80 workers travel from home to work at Office A.

Distance travelled (d km)	Cumulative frequency
$0 < d \leq 5$	17
$0 < d \leq 10$	32
$0 < d \leq 15$	57
$0 < d \leq 20$	70
$0 < d \leq 25$	76
$0 < d \leq 30$	80

- (a) On the grid below, draw a cumulative frequency graph for the information in the table.



(2)

- (b) Use your graph to find an estimate for the median distance travelled.

..... km
(1)

- (c) Use your graph to find an estimate for the interquartile range of the distances travelled.

..... km
(2)

For Office B, the median distance workers travel from home to work is 15 km and the interquartile range is 5 km.

- (d) Use the information above to compare the distances that workers at Office A and workers at Office B travel from home to work.
Write down **two** comparisons.

- 11 The table gives information about the times taken by 90 runners to complete a 10km race.

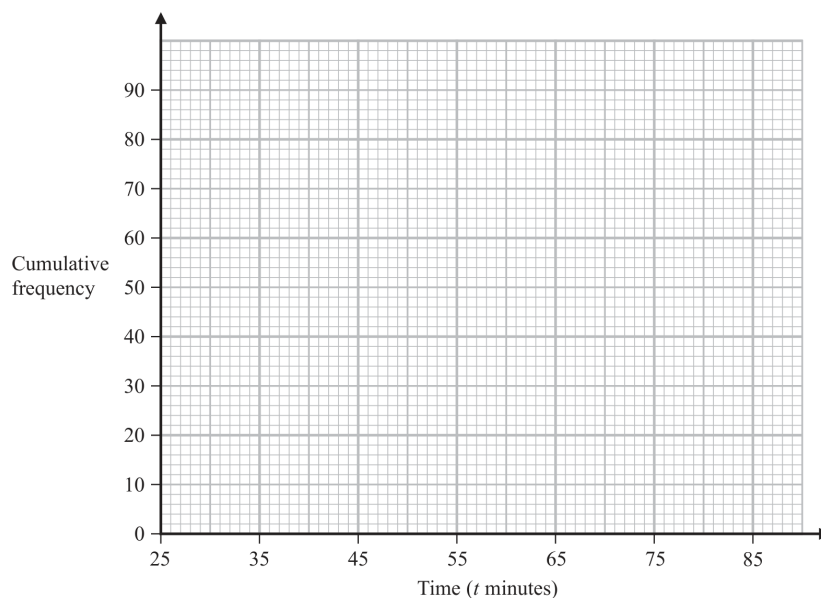
Time (t minutes)	Frequency
$25 < t \leq 35$	12
$35 < t \leq 45$	24
$45 < t \leq 55$	28
$55 < t \leq 65$	12
$65 < t \leq 75$	10
$75 < t \leq 85$	4

- (a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$25 < t \leq 35$	12
$25 < t \leq 45$	
$25 < t \leq 55$	
$25 < t \leq 65$	
$25 < t \leq 75$	
$25 < t \leq 85$	

(1)

- (b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leq 52$ minutes was awarded a silver medal.

- (c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

runners

- 11 The table gives information about the times taken by 90 runners to complete a 10km race.

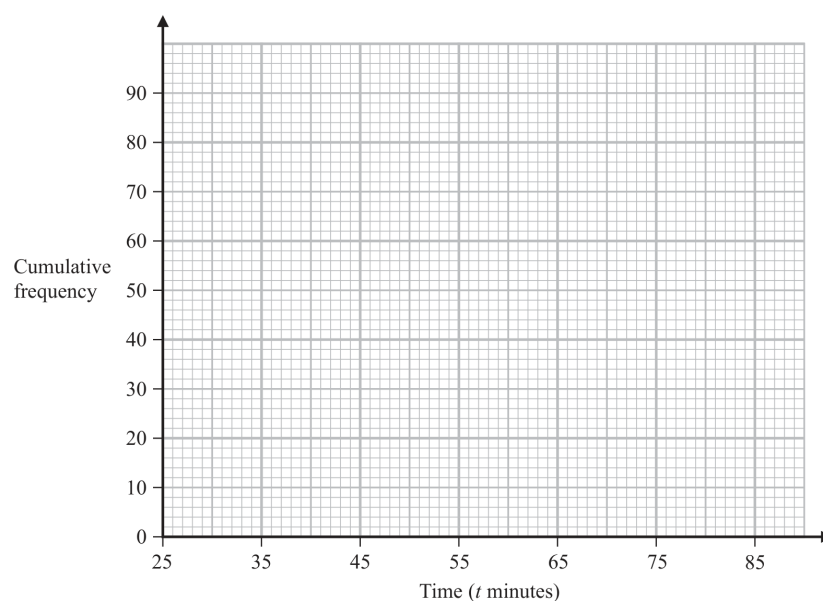
Time (t minutes)	Frequency
$25 < t \leq 35$	12
$35 < t \leq 45$	24
$45 < t \leq 55$	28
$55 < t \leq 65$	12
$65 < t \leq 75$	10
$75 < t \leq 85$	4

- (a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$25 < t \leq 35$	12
$25 < t \leq 45$	
$25 < t \leq 55$	
$25 < t \leq 65$	
$25 < t \leq 75$	
$25 < t \leq 85$	

(1)

- (b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leq 52$ minutes was awarded a silver medal.

- (c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

runners

- 11 The table gives information about the times taken by 90 runners to complete a 10km race.

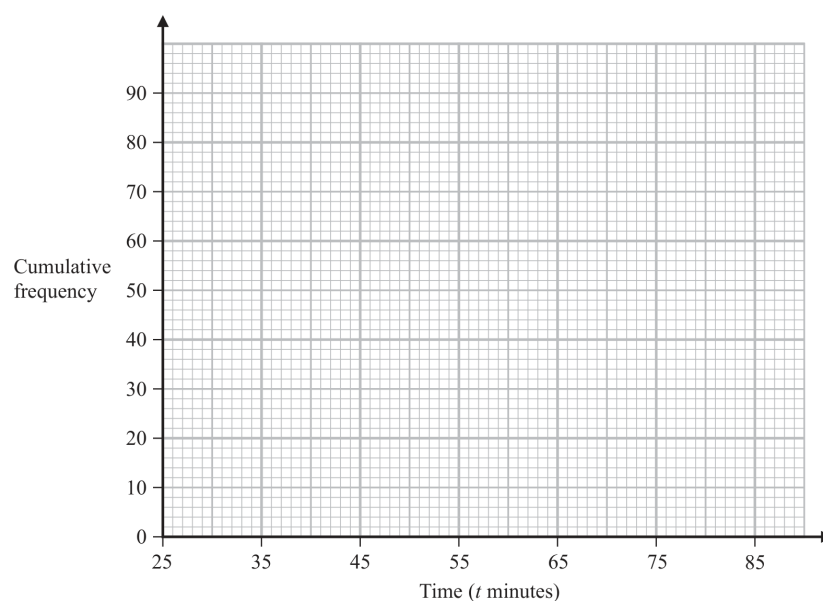
Time (t minutes)	Frequency
$25 < t \leq 35$	12
$35 < t \leq 45$	24
$45 < t \leq 55$	28
$55 < t \leq 65$	12
$65 < t \leq 75$	10
$75 < t \leq 85$	4

- (a) Complete the cumulative frequency table.

Time (t minutes)	Cumulative frequency
$25 < t \leq 35$	12
$25 < t \leq 45$	
$25 < t \leq 55$	
$25 < t \leq 65$	
$25 < t \leq 75$	
$25 < t \leq 85$	

(1)

- (b) On the grid below, draw a cumulative frequency graph for your table.



(2)

Any runner who completed the race in a time T minutes such that $42 < T \leq 52$ minutes was awarded a silver medal.

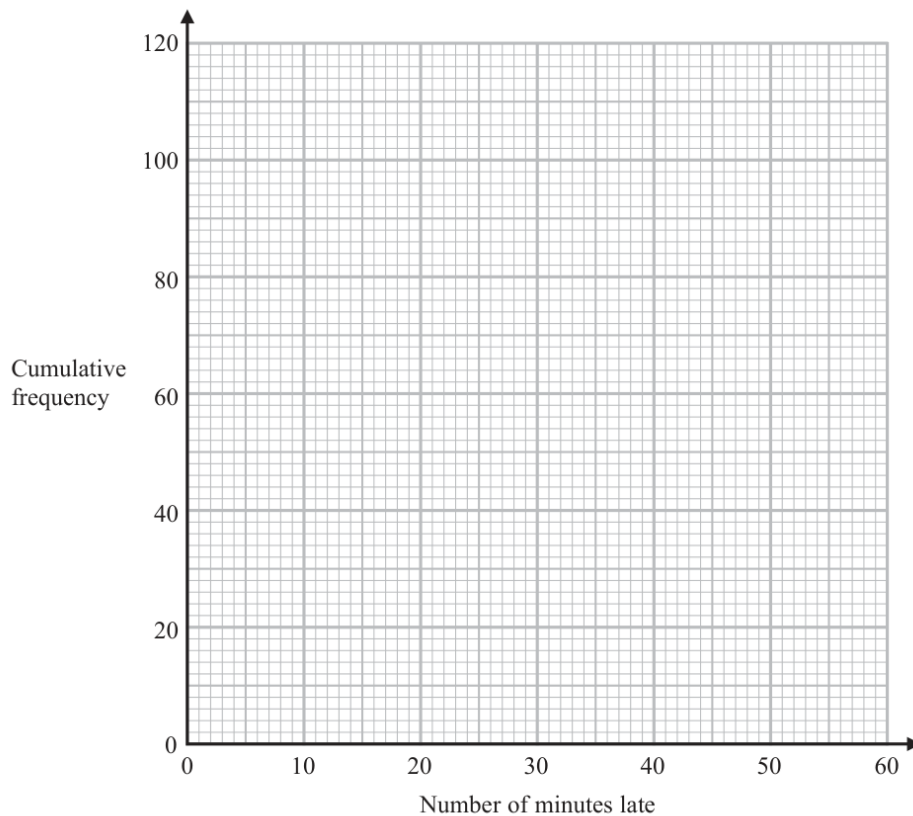
- (c) Use your graph to find an estimate for the number of runners who were awarded a silver medal.

runners

Number of minutes late (L)	Frequency
$0 < L \leq 10$	10
$10 < L \leq 20$	16
$20 < L \leq 30$	44
$30 < L \leq 40$	29
$40 < L \leq 50$	15
$50 < L \leq 60$	6

Number of minutes late (L)	Cumulative frequency
$0 < L \leq 10$	
$0 < L \leq 20$	
$0 < L \leq 30$	
$0 < L \leq 40$	
$0 < L \leq 50$	
$0 < L \leq 60$	

(b) On the grid, draw a cumulative frequency graph for your table.



(2)

(c) Use your graph to find an estimate for the interquartile range.

.....minutes
(2)

(d) Use your graph to find an estimate for the number of buses that were more than 48 minutes late last Monday.

The graph shows the cumulative frequency of a variable over time. The x-axis represents time in minutes, ranging from 0 to 70. The y-axis represents cumulative frequency, ranging from 0 to 60. The curve starts at the origin (0,0) and increases, passing through approximately (10, 3), (20, 8), (30, 15), (40, 25), (50, 40), (60, 55), and ending at (70, 60).

- minutes
(1)

- (b) Use the graph to find an estimate for the number of people who took longer than 55 minutes to shop in the market.

- (c) Use the graph to complete the frequency table to give information about the time, in minutes, each of the 60 people took to shop in the market.

Time taken to shop in the market (m minutes)	Frequency
$0 < m \leq 10$	3
$10 < m \leq 20$	5
$20 < m \leq 30$	
$30 < m \leq 40$	
$40 < m \leq 50$	
$50 < m \leq 60$	
$60 < m \leq 70$	

(2)

(Total for Question 12 is 5 marks)

The graph shows the cumulative frequency of heights. The x-axis is labeled 'Height (cm)' and ranges from 0 to 60 with major grid lines every 10 units and minor grid lines every 2 units. The y-axis is labeled 'Cumulative frequency' and ranges from 0 to 40 with major grid lines every 10 units and minor grid lines every 2 units. The curve starts at the origin (0,0) and follows an S-shape, passing through approximately (10, 4), (20, 11), (30, 24), (40, 34), and ending at (60, 40).

- cm
(1)

- cm
(2)

Plants with a height greater than 40 cm are premium plants.
Greta sells all the premium plants for 30 euros each.

(c) Work out the total amount of money Greta receives for the premium plants.

..... euros

(2)

(Total for Question 14 is 5 marks)